

Generation Interconnection System Impact Study Report

For Queue Project Q14-0314

Sarival 100 kV
October 23, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0314. The Interconnection Customer proposes a 141.2 MW solar photovoltaic facility interconnecting at the Sarival 100 kV substation in the Illinois area. The request was received on March 04, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 175 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,312,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HILLTOP SUBSTATION 500/345kV Transformer Bank	HILLTOP to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500 KV Line	106.4	1,385	7.5
ORCHARD RD to FORT HUMBUG 500 KV	FT HUMBUG to WEST CAMPUS 345 KV	101.5	3,014	12.6
W CAMPUS to FT HUMBUG 345 KV	ORCHARD RD to FT HUMBUG 500 KV	100.8	2,193	8.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HILLTOP 500-345 kV TX	Replace transformer with higher-rated unit	\$26,429,000	2% (\$474,000)
ORCHARD RD to FT HUMBUG 500 KV	Reconductor line	\$23,628,000	2% (\$424,000)
W CAMPUS to FT HUMBUG 345 KV	Reconductor line	\$23,077,000	2% (\$414,000)
	Total allocated to project		\$1,312,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.